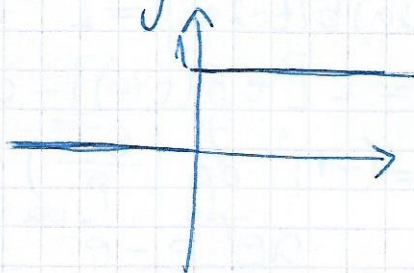


Hevi sajdoma jedinična funkcija

$$\delta(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$

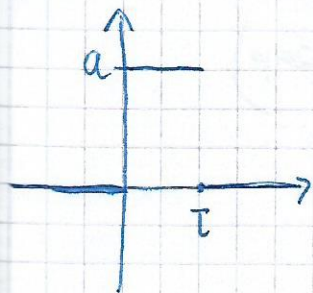


$$L[\delta(t)] = \frac{1}{p}$$

$$\delta(t-\tau) = \begin{cases} 1 & t \geq \tau \\ 0 & t < \tau \end{cases}$$

$$L[\delta(t-\tau)] = \frac{e^{-p\tau}}{p} \quad (\text{osobina } g^\circ)$$

3. Nadi sliku funkcije $f(t) = \begin{cases} a, & 0 < t < \tau \\ 0, & t < 0 \vee t > \tau \end{cases}$



$$f(t) = a [\delta(t) - \delta(t-\tau)]$$

$$L[f(t)] = a (L[\delta(t)] - L[\delta(t-\tau)])$$

$$L[f(t)] = \frac{a}{p} - a \frac{e^{-p\tau}}{p} = \frac{a}{p} (1 - e^{-p\tau})$$

4. Nadi sliku funkcije

$$f(t) = \begin{cases} t, & 0 < t < a \\ 2a - t, & a < t < 2a \\ 0, & t < 0 \vee t \geq 2a \end{cases}$$

$$\begin{aligned} f(t) &= t[\delta(t) - \delta(t-a)] + (2a-t)[\delta(t-a) - \delta(t-2a)] \\ &= t\delta(t) - t\delta(t-a) + (2a-t)\delta(t-a) - (2a-t)\delta(t-2a) \\ &= t\delta(t) + (2a-2t)\delta(t-a) + (t-2a)\delta(t-2a) \\ &= t\delta(t) - 2(t-a)\delta(t-a) + (t-2a)\delta(t-2a) \end{aligned}$$

$$L[\delta(t)] = \frac{1}{p} = F(p)$$

$$L[t\delta(t)] = (-1)^1 \frac{d}{dp} \left(\frac{1}{p} \right) = \frac{1}{p^2}$$

$$L[\delta(t-a)] = \frac{e^{-pa}}{p} = e^{-pa} L[\delta(t)]$$

$$L[(t-a)\delta(t-a)] = L[t\delta(t-a) - a\delta(t-a)] =$$

$$= L[t\delta(t-a)] - aL[\delta(t-a)]$$

$$= (-1)^1 \frac{d}{dp} \left(\frac{e^{-pa}}{p} \right) - a \cdot \frac{e^{-pa}}{p}$$

$$= - \frac{-ae^{-pa} \cdot p - e^{-pa}}{p^2} - a \cdot \frac{e^{-pa}}{p} = \frac{(-ap e^{-pa} - e^{-pa}) + ap e^{-pa}}{p^2}$$

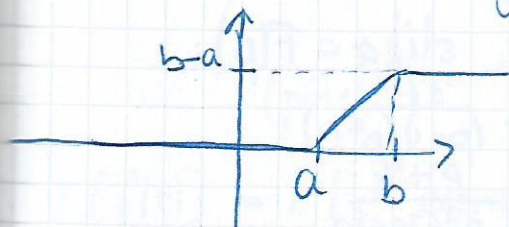
$$= \frac{ap e^{-pa} + e^{-pa} - ap e^{-pa}}{p^2} = \frac{e^{-pa}}{p^2}$$

$$L[\delta(t-2a)] = e^{-p2a} \cdot \frac{1}{p} = \frac{e^{-2ap}}{p}$$

$$L[(t-2a)\delta(t-2a)] = \frac{e^{-2ap}}{p^2}$$

$$F(p) = \frac{1}{p^2} - 2 \frac{e^{-ap}}{p^2} + \frac{e^{-2ap}}{p^2} = \frac{1}{p^2} (1 - 2e^{-ap} + e^{-2ap})$$

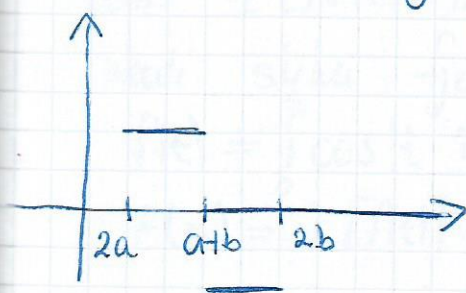
5. Nadi sliku funkcije $f(t) = \begin{cases} 0, & t < a \\ t-a, & a < t < b \\ b-a, & t > b \end{cases}$



$$\begin{aligned} f(t) &= (t-a)[\delta(t-a) - \delta(t-b)] + (b-a)\delta(t-b) \\ &= (t-a)\delta(t-a) - (t-a)\delta(t-b) + (b-a)\delta(t-b) \\ &= (t-a)\delta(t-a) - (t-a-b+a)\delta(t-b) \\ &= (t-a)\delta(t-a) - (t-b)\delta(t-b) \end{aligned}$$

$$\begin{aligned} F(p) &= L[(t-a)\delta(t-a)] - L[(t-b)\delta(t-b)] \\ &= \frac{e^{-pa}}{p^2} - \frac{e^{-pb}}{p^2} = \frac{1}{p^2} (e^{-ap} - e^{-bp}) \end{aligned}$$

6. Nadi sliku fje $f(t) = \begin{cases} 1, & -2a < t < a+b \\ -1, & a+b < t < 2b \\ 0, & t < -2a \vee t > 2b \end{cases}$



$$\begin{aligned} f(t) &= \delta(t+2a) - \delta(t-a-b) - \delta(t-a-b) + \delta(t-2b) \\ &= \delta(t+2a) - 2\delta(t-a-b) + \delta(t-2b) \\ &= \frac{e^{-2ap}}{p} - 2\frac{e^{-(a+b)p}}{p} + \frac{e^{-2bp}}{p} \\ &= \frac{1}{p} (e^{-2ap} - 2e^{-ap}e^{-bp} + e^{-2bp}) = \frac{1}{p} (e^{-ap} - e^{-bp})^2 \end{aligned}$$

Nalazhenje originala

1. Nadi original $f(t)$, ako je poznata slika $F(p)$:

a) $F(p) = \frac{2}{p} + \frac{2}{p^2+16}$

c) $F(p) = \frac{1}{(p-1)(p^2+1)}$

b) $F(p) = \frac{1}{p^2+2p+5}$

d) $F(p) = \frac{p+3}{p^2+2p+10}$

a) $F(p) = \frac{2}{p} + \frac{2}{p^2+16} = 2 \cdot \frac{1}{p} + \frac{1}{2} \cdot \frac{4}{p^2+16}$

$L[f(t)] = 2 L[6(t)] + \frac{1}{2} L[\sin 4t]$

$\Rightarrow f(t) = 2 \sin 4t + \frac{1}{2} \sin 4t$

b) $F(p) = \frac{1}{p^2+2p+5} = \frac{1}{p^2+2p+1+4} = \frac{1}{(p+1)^2+4} = \frac{1}{2} \cdot \frac{2}{(p+1)^2+2^2}$

$L[f(t)] = \frac{1}{2} L(e^{-t} \sin 2t)$

$f(t) = \frac{1}{2} e^{-t} \sin 2t$

c) $F(p) = \frac{1}{(p-1)(p^2+1)} = \frac{A}{p-1} + \frac{Bp+C}{p^2+1}$

$1 = A(p^2+1) + (Bp+C)(p-1)$

$1 = Ap^2 + A + Bp^2 - Bp + Cp - C$

$A+B=0$

$-B+C=0 \Rightarrow B=C$

$A-C=1$

$A+B=0$

$A-B=1$

$2A=1$

$A=\frac{1}{2}$

$B=-A$

$B=-\frac{1}{2}$

$C=-\frac{1}{2}$

$F(p) = \frac{1}{2} \frac{1}{p-1} + \frac{-\frac{1}{2}p - \frac{1}{2}}{p^2+1} = \frac{1}{2} \cdot \frac{1}{p-1} - \frac{1}{2} \cdot \frac{p+1}{p^2+1} =$

$= \frac{1}{2} \cdot \frac{1}{p-1} - \frac{1}{2} \frac{p}{p^2+1} - \frac{1}{2} \frac{1}{p^2+1}$

$L[f(t)] = \frac{1}{2} L[e^t] - \frac{1}{2} L[\cos t] - \frac{1}{2} L[\sin t]$

$f(t) = \frac{1}{2} (e^t - \cos t - \sin t)$

$$d) F(p) = \frac{p+3}{p^2+2p+10} = \frac{p+3}{(p+1)^2+9} = \frac{p+1}{(p+1)^2+3^2} + \frac{2}{(p+1)^2+3^2} =$$

$$= \frac{p+1}{(p+1)^2+3^2} + \frac{2}{3} \cdot \frac{3}{(p+1)^2+3^2}$$

$$L[f(t)] = L[e^{-t} \cos 3t] + \frac{2}{3} L[e^{-t} \sin 3t]$$

$$f(t) = e^{-t} \cos 3t + \frac{2}{3} e^{-t} \sin 3t$$

Konvolucija

Def: Konvolucija neprekidnih funkcija f_1 i f_2 u oznaci $f_1 * f_2$ je funkcija definisana formulom

$$f(t) = f_1(t) * f_2(t) = \int_0^t f_1(t-\tau) f_2(\tau) d\tau$$

Teorem: $L[f_1(t) * f_2(t)] = L[f_1(t)] \cdot L[f_2(t)] = F_1(p) \cdot F_2(p)$

1. Nadi sliku fje $f(t)$ ako je

$$f(t) = \int_0^t \cos(t-\tau) \sin \tau d\tau = \cos t * \sin t$$

$$F(p) = L[f(t)] = L[\cos t * \sin t] = L[\cos t] \cdot L[\sin t] =$$

$$= \frac{p}{p^2+1} \cdot \frac{1}{p^2+1} = \frac{p}{(p^2+1)^2}$$

$$2. f(t) = \int_0^t e^{t-\tau} \cos 2(t-\tau) \sin \tau d\tau = e^t \cos 2t * \sin t$$

$$F(p) = L[e^t \cos 2t] L[\sin t] = \frac{p-1}{(p-1)^2+4} \cdot \frac{1}{p^2+1} = \frac{1}{(p+1)((p-1)^2+4)}$$

3. Koristeci teorem o konvoluciji nadi original

$$a) F(p) = \frac{p}{(p^2+4)^2}$$

$$b) F(p) = \frac{1}{p^2(p-1)}$$

$$c) F(p) = \frac{p}{(p-1)(p^2+1)}$$

$$\text{rez: } f(t) = e^t - t - 1$$

$$\text{rez: } \frac{1}{2} (\sin t - \cos t + e^t)$$

$$a) F(p) = \frac{P}{(p^2+4)^2} = \frac{P}{p^2+4} \cdot \frac{1}{p^2+4} = \frac{1}{2} \frac{P}{p^2+4} \cdot \frac{2}{p^2+4}$$

$$F(p) = \frac{1}{2} L[\cos 2t] \cdot L[\sin 2t] = \frac{1}{2} L[\cos 2t * \sin 2t]$$

$$f(t) = \frac{1}{2} \int_0^t \cos 2(t-\tau) \sin 2\tau d\tau$$

$$+ \sin x \sin y = \frac{1}{2} (\cos(x-y) - \cos(x+y)) +$$

$$\sin x \cos y = \frac{1}{2} (\sin(x-y) + \sin(x+y))$$

$$+ \cos x \cos y = \frac{1}{2} (\cos(x-y) + \cos(x+y)) +$$

$$f(t) = \frac{1}{2} \int_0^t \sin 2\tau \cos(2t-2\tau) d\tau$$

$$= \frac{1}{2} \int_0^t \frac{1}{2} (\sin(2\tau-2t+2\tau) + \sin(2t+2\tau-2\tau)) d\tau$$

$$= \frac{1}{4} \int_0^t (\sin(4\tau-2t) + \sin 2t) d\tau$$

$$= \frac{1}{4} \int_0^t \sin(4\tau-2t) d\tau + \frac{1}{4} \int_0^t \sin 2t d\tau = \left| \begin{array}{l} 4\tau-2t=z \\ 4d\tau=dz \end{array} \right|$$

$$= \frac{1}{4} \cdot \int_{(0)}^{(t)} \sin z \frac{1}{4} dz + \frac{1}{4} \sin 2t \cdot \tau \Big|_0^t =$$

$$= \frac{1}{16} \cos z \Big|_{(0)}^{(t)} + \frac{1}{4} t \sin 2t =$$

$$= \frac{1}{4} t \sin 2t - \frac{1}{16} \cos(4\tau-2t) \Big|_0^t = \frac{1}{4} t \sin 2t - \frac{1}{16} (\cos 2t - \cos(-2t))$$

$$= \frac{1}{4} t \sin 2t$$

$$4. a) F(p) = \frac{P}{p^2+9} \cdot \frac{P}{p^2+9} = L[\cos 3t] \cdot L[\cos 3t]$$

$$f(t) = f_1(t) * f_2(t) = \cos 3t * \cos 3t = \int_0^t \cos(3t-3\tau) \cdot \cos 3\tau d\tau =$$

$$= \frac{1}{2} \int_0^t (\cos(3t-3\tau-3\tau) + \cos(3t-3\tau+3\tau)) d\tau$$

$$= \frac{1}{2} \int_0^t (\cos(3t-6\tau) + \cos 3t) d\tau$$

$$= \frac{1}{2} \int_0^t \cos(3t-6\tau) d\tau + \frac{1}{2} \int_0^t \cos 3t d\tau$$

$$= \frac{1}{2} t \cos 3t + \frac{1}{2} \int_0^t \cos(3t-6\tau) d\tau = \left| \begin{array}{l} 3t-6\tau=z \\ -6d\tau=dz \\ d\tau = -\frac{1}{6} dz \end{array} \right|$$

$$= \frac{1}{2} t \cos 3t - \frac{1}{2} \cdot \frac{1}{6} \int_0^t \cos z \, dz = \frac{1}{2} t \cos 3t - \frac{1}{12} \sin(3t - 6t) \Big|_0^t$$

$$= \frac{1}{2} t \cos 3t - \frac{1}{12} (\sin(-3t) - \sin 3t) = \frac{1}{2} t \cos 3t + \frac{1}{6} \sin 3t$$

$$b) F(p) = \frac{1}{p(p^2 - 4p + 5)}$$

$$\text{Stavimo } F_1(p) = \frac{1}{p^2 - 4p + 5} = \frac{1}{(p-2)^2 + 1} = L[e^{-2t} \sin t]$$

$$F(p) = \frac{F_1(p)}{p}$$

$$\gamma \Rightarrow \frac{F_1(p)}{p} = L\left[\int_0^t e^{2u} \sin u \, du\right] = L\left[\frac{-\cos u e^{2u} + 2e^{2u} \sin u}{5} \Big|_0^t\right]$$

$$\frac{1}{p(p^2 - 4p + 5)} = L\left[\frac{-\cos t e^{2t} + 2e^{2t} \sin t}{5} + \frac{1}{5}\right]$$

$$f(t) = \frac{1}{5} (-\cos t e^{2t} + 2e^{2t} \sin t + 1)$$

Rješavanje diferencijalnih jednačina

1. Primjenom Lapl. transformacije riješiti dif. jednačine

$$a) \quad x''(t) + 4x(t) = 2 \quad x(0) = x'(0) = 0$$

$$L[x(t)] = X(p)$$

$$L[x''(t)] = p^2 X(p) + p x(0) - x'(0) = p^2 X(p)$$

$$\Rightarrow x'(t) + 4x(t) = 2 \quad / \quad L[\quad]$$

$$L[x''(t)] + 4L[x(t)] = 2L[1]$$

$$p^2 X(p) + 4X(p) = \frac{2}{p}$$

$$X(p)(p^2 + 4) = \frac{2}{p}$$

$$X(p) = \frac{2}{p(p^2 + 4)}$$

$$\frac{2}{p(p^2 + 4)} = \frac{A}{p} + \frac{Bp + C}{p^2 + 4}$$

$$A = \frac{1}{2} \quad B = -\frac{1}{2} \quad C = 0$$

$$X(p) = \frac{1}{2} \left(\frac{1}{p} - \frac{p}{p^2 + 4} \right)$$

$$x(t) = \frac{1}{2} L^{-1} \left(\frac{1}{p} - \frac{p}{p^2 + 4} \right)$$

$$x(t) = \frac{1}{2} (1 - \cos 2t)$$

$$b) \quad y''(t) - 2y'(t) + 2y(t) = 2e^t \cos t \quad y(0) = y'(0) = 0$$

$$\mathcal{L}[y(t)] = Y(p)$$

$$\mathcal{L}[y'(t)] = pY(p) - y(0) = pY(p)$$

$$\mathcal{L}[y''(t)] = p^2 Y(p) - py(0) - y'(0) = p^2 Y(p)$$

$$\mathcal{L}[2e^t \cos t] = 2 \mathcal{L}[e^t \cos t] = 2 \cdot \frac{p-1}{(p-1)^2 + 1}$$

$$\Rightarrow p^2 Y(p) - 2pY(p) + 2Y(p) = 2 \frac{p-1}{(p-1)^2 + 1}$$

$$(p^2 - 2p + 2) Y(p) = 2 \frac{p-1}{(p-1)^2 + 1}$$

$$Y(p) = 2 \frac{p-1}{[(p-1)^2 + 1](p^2 - 2p + 2)}$$

$$Y(p) = \underbrace{\frac{1}{(p-1)^2 + 1}}_{F_1(p)} \cdot \underbrace{\frac{2(p-1)}{(p-1)^2 + 1}}_{F_2(p)}$$

$$f_1(t) = \mathcal{L}^{-1}[F_1(p)] = \mathcal{L}^{-1}\left[\frac{1}{(p-1)^2 + 1}\right] = e^t \sin t$$

$$f_2(t) = \mathcal{L}^{-1}[F_2(p)] = \mathcal{L}^{-1}\left[\frac{2(p-1)}{(p-1)^2 + 1}\right] = 2e^t \cos t$$

$$y(t) = f_1(t) * f_2(t) = \int_0^t f_1(\tau) \cdot f_2(t-\tau) d\tau$$

$$y(t) = 2 \int_0^t e^{\tau} \sin \tau \cdot e^{t-\tau} \cos(t-\tau) d\tau$$

$$y(t) = 2e^t \int_0^t \sin \tau \cos(t-\tau) d\tau$$

$$y(t) = 2e^t \cdot \frac{1}{2} \int_0^t [\sin(\tau+t-\tau) + \sin(\tau-t+\tau)] d\tau$$

$$y(t) = e^t \int_0^t (\sin t + \sin(2\tau-t)) d\tau$$

$$y(t) = e^t \left(t \sin t + \int_0^t \sin(2\tau-t) d\tau \right) \quad \left| \begin{array}{l} 2\tau-t = u \\ 2d\tau = du \\ d\tau = \frac{du}{2} \end{array} \right. \quad \tau \Big|_0^t \Rightarrow u \Big|_{-t}^t$$

$$y(t) = e^t \left(t \sin t + \frac{1}{2} \int_{-t}^t \sin u du \right)$$

$$y(t) = e^t \left(t \sin t - \frac{1}{2} \cos u \Big|_{-t}^t \right) = e^t \left(t \sin t - \frac{1}{2} (\cos t - \cos(-t)) \right)$$

$$y(t) = te^t \sin t$$

$$c) \quad y''(t) + y(t) = \sin t \quad y(0) = 0 \quad y'(0) = -\frac{1}{2}$$

$$\mathcal{L}[y(t)] = Y(p)$$

$$\mathcal{L}[y''(t)] = p^2 Y(p) - p y(0) - y'(0) = p^2 Y(p) + \frac{1}{2}$$

$$\mathcal{L}[\sin t] = \frac{1}{p^2 + 1}$$

$$\Rightarrow p^2 Y(p) + \frac{1}{2} + Y(p) = \frac{1}{p^2 + 1}$$

$$Y(p)(p^2 + 1) = \frac{1}{p^2 + 1} - \frac{1}{2}$$

$$(p^2 + 1) Y(p) = \frac{2 - p^2 - 1}{2(p^2 + 1)} = \frac{1 - p^2}{2(p^2 + 1)}$$

$$Y(p) = \frac{1 - p^2}{2(p^2 + 1)^2}$$

$$y(t) = \mathcal{L}^{-1}\left[\frac{1 - p^2}{2(p^2 + 1)^2}\right] = \mathcal{L}^{-1}\left[-\frac{p^2 - 1}{2(p^2 + 1)^2}\right] = \mathcal{L}^{-1}\left[-\frac{p^2 + 1 - 2}{2(p^2 + 1)^2}\right]$$

$$= \mathcal{L}^{-1}\left[-\frac{p^2 + 1}{2(p^2 + 1)^2} + \frac{1}{(p^2 + 1)^2}\right]$$

$$= \mathcal{L}^{-1}\left[-\frac{1}{2(p^2 + 1)}\right] + \mathcal{L}^{-1}\left[\frac{1}{(p^2 + 1)^2}\right]$$

$$y(t) = -\frac{1}{2} \sin t + \mathcal{L}^{-1}\left[\frac{1}{p^2 + 1} \cdot \frac{1}{p^2 + 1}\right]$$

$$y(t) = -\frac{1}{2} \sin t + \int_0^t \sin \tau \sin(t - \tau) d\tau$$

$$y(t) = -\frac{1}{2} \sin t + \frac{1}{2} \int_0^t (\cos(\tau - t + \tau) - \cos(\tau + t - \tau)) d\tau$$

$$y(t) = -\frac{1}{2} \sin t + \frac{1}{2} \int_0^t [\cos(2\tau - t) - \cos t] d\tau$$

$$y(t) = -\frac{1}{2} \sin t + \frac{1}{2} \int_0^t \cos(2\tau - t) d\tau - \frac{1}{2} \cos t \cdot t$$

$$\begin{aligned} 2\tau - t &= u \\ d\tau &= \frac{1}{2} du \\ \tau \Big|_0^t &\Rightarrow u \Big|_{-t}^t \end{aligned}$$

$$y(t) = -\frac{1}{2} \sin t - \frac{1}{2} t \cos t + \frac{1}{4} \sin(2\tau - t) \Big|_0^t$$

$$y(t) = -\frac{1}{2} \sin t - \frac{1}{2} t \cos t + \frac{1}{4} (\underbrace{\sin t - \sin(-t)}_{= 2 \sin t})$$

$$y(t) = \cancel{-\frac{1}{2} \sin t} - \frac{1}{2} t \cos t + \cancel{\frac{1}{2} \sin t}$$

$$y(t) = -\frac{1}{2} t \cos t$$

$$d) \quad y'' - 3y' + 2y = 4t + 12e^{-t} \quad y(0) = 0 \quad y'(0) = 7$$

$$\text{rez: } y(t) = 3 + 2t - 17e^t + 12e^{2t} + 2e^{-t}$$

$$e) \quad y'' + 2y' + y = \sin t \quad y(0) = y'(0) = y''(0) = y'''(0) = 0$$